Distance Meter (Basic)

Arduino Uno R3 + Tinkercad | HC-SR04 Ultrasonic Sensor — Serial Monitor Output

Yeh project HC-SR04 ultrasonic sensor use karke distance measure karta hai aur result seedha **Serial Monitor** par cm mein dikhata hai. Code is PDF mein copy-paste ke liye maujood hai, aur sath mein alag **Distance_Meter_Basic.ino** file bhi di gayi hai jo seedha Tinkercad ya Arduino IDE mein open ki ja sakti hai.

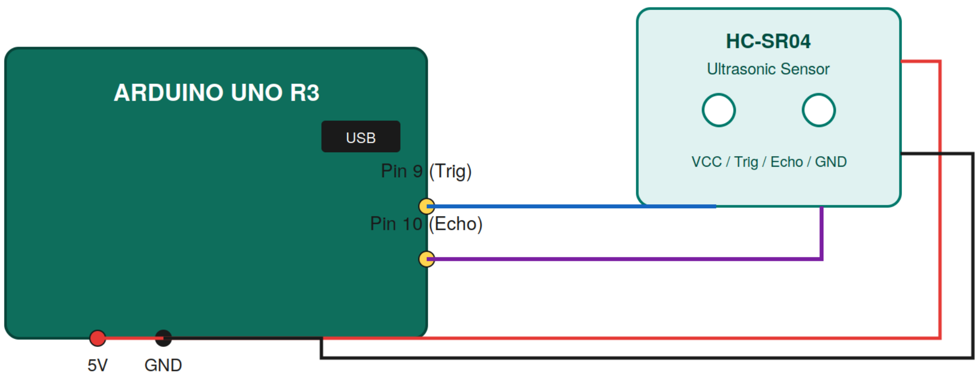
1. Components Used

Component	Quantity	Purpose
Arduino Uno R3	1	Main microcontroller board
HC-SR04 Ultrasonic Sensor	1	Measures distance using sound waves
Breadboard	1	For connections
Jumper Wires	As needed	For wiring all components

2. Circuit Diagram

White background par color-coded wiring niche di gayi hai (legend diagram ke neeche hai):

Distance Meter (Basic) — Wiring Diagram



Legend: 5V (Power) GND Trig (Output) Echo (Input)

Note: HC-SR04 VCC connects to Arduino 5V and GND connects to Arduino GND.

3. Build Steps

1. Tinkercad open karo aur naya circuit banao.
2. Arduino Uno R3 aur breadboard components search karke drag karo.
3. HC-SR04 ultrasonic sensor bhi search karke breadboard pe rakho.
4. Wiring karo: HC-SR04 VCC to Arduino 5V, GND to Arduino GND, Trig to Arduino Pin 9, Echo to Arduino Pin 10.
5. Code editor open karo aur niche diya gaya code paste karo.
6. Simulation start karo aur Serial Monitor open karke distance values dekho (cm mein).

4. Full Arduino Code

```
#define trigPin 9
#define echoPin 10

void setup() {
  Serial.begin(9600);
  pinMode(trigPin, OUTPUT);
  pinMode(echoPin, INPUT);
}

void loop() {
  long duration, distance;

  digitalWrite(trigPin, LOW);
  delayMicroseconds(2);
  digitalWrite(trigPin, HIGH);
  delayMicroseconds(10);
  digitalWrite(trigPin, LOW);

  duration = pulseIn(echoPin, HIGH);
  distance = duration * 0.034 / 2;

  Serial.print("Distance: ");
  Serial.print(distance);
  Serial.println(" cm");

  delay(500);
}
```

Tip: Is code ko seedha copy-paste kar sakte ho, ya sath diya gaya **Distance_Meter_Basic.ino** file directly Tinkercad/Arduino IDE mein open kar lo.

5. Wiring Summary Table

Component Pin	Arduino Pin	Pin Type
HC-SR04 - VCC	5V	Power
HC-SR04 - GND	GND	Ground
HC-SR04 - Trig	Pin 9	Digital Output
HC-SR04 - Echo	Pin 10	Digital Input

6. Kaise Kaam Karta Hai

- Arduino Trig pin (9) se 10 microseconds ka HIGH pulse bhejta hai, jisse HC-SR04 se ek ultrasonic (40kHz) sound wave nikalti hai.
- Yeh sound wave saamne wali cheez se takra kar wapas aati hai, aur sensor apne Echo pin (10) ko us waqt tak HIGH rakhta hai jab tak echo wapis nahi aati.
- Arduino **pulseIn()** function se yeh maapta hai ke Echo pin kitni der HIGH raha (microseconds mein) — yeh 'duration' hai.
- Distance nikalne ka formula: **distance = duration * 0.034 / 2**. Sound ki speed ~0.034 cm/microsecond hoti hai, aur 2 se divide isliye karte hain kyunki wave jaake wapas aati hai (round trip).
- Har 500 milliseconds baad yeh process loop() mein dobara chalta hai aur naya distance value Serial Monitor par print hoti hai.